

## Multiple Choice (10 marks)

1. During the manic phase of a bipolar disorder, individuals are most likely to experience
  - (a) high self-esteem
  - (b) delusions of persecution
  - (c) uncontrollable grief and despair
  - (d) visual hallucinations
2. The main reason given by men and women for their long lasting marriage is that
  - (a) They are best friends
  - (b) They must fulfill their vows
  - (c) They agree on almost everything
  - (d) They owe it to their children
3. Which of the following is (are) required when performing a polymerase chain reaction (P C R) on a DNA sample?
  - (a) Reverse transcriptase
  - (b) A vacuum chamber
  - (c) Primers complementary to certain portions of the DNA
  - (d) DNA ligase
4. Which of the following causes a chemical change?
  - (a) sugar added to soda
  - (b) detergent added to water
  - (c) lemon juice added to milk
  - (d) salt added to water
5. Which of the following is NOT a way in which the powers of the executive branch are balanced by the legislative branch?
  - (a) The Senate may use the filibuster to block an executive order.
  - (b) Congress may impeach and convict a president for "high crime and misdemeanors."
  - (c) The Senate can decline to ratify a treaty.
  - (d) The Senate can decline to approve a president's nominee for the Supreme Court.

## True / False (10 marks)

*Answer True or False and justify briefly.*

1. True or False: The correct answer to 'which of these would more likely survive?' is 'a lost dog by the river, with small rodents'.
2. True or False: The correct answer to 'A creature's habitat would be like what, if it has wide appendages that grasp liquids well?' is 'salt water'.
3. True or False: The correct answer to 'Imagine an organism whose  $2n = 96$ . Meiosis would leave this organism's cells with how many chromosomes?' is '48'.
4. True or False: The correct answer to 'How much toothpaste is recommended for cleaning teeth?' is 'The length of the toothbrush.'.
5. True or False: The correct answer to 'Earth's rotation (turning on its axis) causes' is 'day and night.'.

## Long Form Response (20 marks)

*Provide thorough reasoning and reference key steps.*

1. Write a python function to identify non-prime numbers.
2. Write a function to find the  $n$ 'th lucas number.

*End of Paper*